

Master Guide — Phase Portraits and Orbit Shapes

1. Straight Lines

Usually appear for nodes or invariant eigenvector directions. Typical equations: $y = cx$. Arrows: e^{-t} means toward the origin, e^t means away from the origin.

2. Hyperbolas

Usually appear for saddles. Typical first integrals: $xy = c$ or $y^2 - ax^2 = c$. Draw two invariant lines and hyperbola-like curves between them. One invariant line goes inward, the other outward.

3. Circles

Usually correspond to centers. Conditions: purely imaginary eigenvalues or $\rho' = 0$ in polar coordinates. Direction determined by θ' : positive means counterclockwise, negative means clockwise.

4. Ellipses

Also correspond to centers. Typical first integrals: $ax^2 + by^2 = c$. Direction again determined by θ' .

5. Spirals

Appear when eigenvalues are complex with nonzero real part: $a \pm bi$. If $a < 0$: spiral inward (stable focus). If $a > 0$: spiral outward (unstable focus).

6. Exponential Curves

Usually in scalar ODEs $x' = ax$. Solutions: $x(t) = Ce^{at}$. If $a > 0$: exponential growth. If $a < 0$: exponential decay.

7. Random Curves

For nonlinear systems, use equilibria, signs, nullclines, asymptotes, and invariant regions. You sketch topology, not exact formulas.

8. How to Draw Arrows

For scalar ODEs $x' = f(x)$: $f(x) > 0$ gives arrows right, $f(x) < 0$ gives arrows left. For planar systems: use eigenvectors/eigenvalues, test points, or polar coordinates.

9. Fast Exam Template

Real eigenvalues same sign $\rightarrow$ node. Opposite signs $\rightarrow$ saddle. Purely imaginary $\rightarrow$ center. Complex with real part $\rightarrow$ focus.

10. Shape Recognition Table

Equation / Integral	Shape
$y = cx$	Lines
$xy = c$	Hyperbolas
$x^2 + y^2 = c$	Circles
$ax^2 + by^2 = c$	Ellipses
Complex eigenvalues	Spirals
Ce^{at}	Exponential graph